

Homework 1 Due Friday, January 30, 2026 at 6pm (via Gradescope.com)**Problem 1** Text problem 3.6.**Problem 2** Text problem 3.9. In part b), make sure to use part a) to establish that V is radially unbounded.**Problem 3** Text problem 3.11.**Problem 4** Text problem 3.15.**Problem 5** Text problem 3.18.**Problem 6** Text problem 3.20.**Problem 7** Consider the differential equation $\dot{x} = f(x)$ of the form

$$\begin{aligned}\dot{x}_1 &= 2x_2 - \frac{6x_1}{(1+x_1^2)^2} \\ \dot{x}_2 &= \frac{-2(x_1+x_2)}{(1+x_1^2)^2}.\end{aligned}$$

Show, for the function $V(x) := \frac{x_1^2}{1+x_1^2} + x_2^2$ (as in Figure 3.7), that $\langle \nabla V(x), f(x) \rangle < 0$ for all $x \in \mathbb{R}^2 \setminus \{0\}$. It can be shown (you do not need to do this) that solutions starting to the right of the branch of the hyperbola $x_2 = 2(x_1 - \sqrt{2})$ in the first quadrant cannot cross that branch. Using Simulink, simulate the solution from points of the form $(x_1(0), x_2(0)) = (2\sqrt{2}, \sqrt{2})$ to verify this property, and comment on whether the origin is GAS.

Problem 8 Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable and $A \in \mathbb{R}^{m \times n}$. Suppose that the point $(x^*, \nu^*) \in \mathbb{R}^n \times \mathbb{R}^m$ satisfies the conditions

$$\begin{aligned}0 &= \nabla f(x^*) + A^T \nu^* \\ 0 &= Ax^* - b\end{aligned}$$

Show that the point (x^*, ν^*) is globally asymptotically stable (GAS) for the system

$$\begin{aligned}\dot{x} &= -\nabla f(x) - A^T \nu \\ \dot{\nu} &= Ax - b\end{aligned}$$

when A has rank m and f is strictly convex, namely

$$f(y) > f(x) + \langle \nabla f(x), y - x \rangle$$

for all $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n$ such that $y \neq x$. Establish GAS first using a weak Lyapunov function and showing that no complete solution keeps the Lyapunov function equal to a nonzero constant. Hint: the simplest Lyapunov function you can imagine should work.